

Luke Caffarey

Raleigh, NC | 919-695-6651 | lacaffar@ncsu.edu
linkedin.com/in/luke-caffarey-212473322 | github.com/lacaffar | lacaffar.github.io

Mechanical engineering student with hands-on experience across fusion research and utility-scale wind, now building custom rocket avionics and remote telemetry hardware. Drawn to spaceflight and satellite systems: hardware that has to work the first time, far from the people who built it.

EDUCATION

North Carolina State University | Raleigh, NC

- B.S. Mechanical Engineering | Expected May 2028 | GPA: 3.69
- Relevant tools and coursework: SolidWorks, MATLAB, Python, thermodynamics, fluid mechanics, mechanics of materials.

EXPERIENCE

GE Vernova, Wind Engineer Intern | May 2026 to Present | Greenville, SC

- Support engineering on utility-scale wind turbine hardware within GE Vernova's onshore wind business.
- Automated a weekly safety-critical reporting workflow in Python and SQL, replacing a manual multi-step process.
- Apply CAD and mechanical analysis to hardware that must operate reliably in the field for decades.

NC State Mechanical Engineering, Senior Design Teaching Assistant | Aug 2026 to Present | Raleigh, NC

- Will support senior capstone design teams through design reviews, prototyping guidance, and documentation feedback.

FPAC Laboratory, NC State, Undergraduate Researcher | Dec 2024 to Apr 2025 | Raleigh, NC

- Analyzed spectroscopy and video data from KSTAR fusion reactor experiments to support reactor wall stability.
- Built Python workflows for time-series signal processing, using adaptive Gaussian smoothing to improve peak detection.

LEADERSHIP AND INVOLVEMENT

Windpack Club, NC State, Founder and President | Aug 2025 to Present | Raleigh, NC

- Founded NC State's Collegiate Wind Competition team and grew it to 25+ members across three sub-teams (Mechanical, Electrical, Project Development), leading it to its first national competition appearance in 2026, placing 6th out of 10.
- Lead aerodynamic design of the competition rotor in QBlade, running airfoil trade studies to produce power across 5-11 m/s winds and survive 13 m/s durability testing within a 45 cm rotor envelope.
- Directing design and wind-tunnel testing of three blade variants (maximum performance, balanced, and composite-manufacturable) against competition power-curve scoring.
- Piloting a composite blade layup process alongside the club's standard 3D-printed blades, recruiting NC State faculty mentorship for composite manufacturing.
- Oversee the project development contest entry: wind resource assessment, site selection, environmental review, and grid interconnection for a full wind farm proposal.
- Built the club's engineering infrastructure and pipeline: Siemens NX CAD on a VPN-accessed club server, an organized officer structure, and scoped summer projects (generator dynamometer, rotor test stand) that train new members.

PROJECTS

High-Power Rocket Avionics and Remote Ground Station (in progress) | 2026

- Designing a custom flight computer (Teensy 4.1, IMU, barometer, GPS) that logs 100 Hz flight data and streams live telemetry over a 915 MHz LoRa radio link to a Raspberry Pi ground station with a real-time web dashboard.
- Flight-testing the telemetry stack on a quadcopter to characterize radio dropouts, range limits, and antenna placement before it flies on a high-power rocket.

ADDITIONAL EXPERIENCE

Annelore's German Bakery, Assistant | May 2025 to May 2026 | Cary, NC

- Support weekend operations (16 hrs/week): baking assistance, order taking, and food prep.

Wake County Government, EMS Division, EMS Cadet | Jan 2022 to May 2024 | Raleigh, NC

- Shadowed paramedics in 911 emergency response, assisting with patient care in high-pressure conditions.

SKILLS

CAD and Design: SolidWorks, Siemens NX, QBlade, design for manufacturing, 3D printing

Programming and Data: Python (signal processing, data analysis), MATLAB, SQL, C++ (embedded)

Hardware: Fabrication, soldering, composite layup, sensor integration (IMU, barometer, GPS), LoRa radio telemetry

Systems: Raspberry Pi and Linux, Git, remote networking (SSH, Tailscale), thermodynamics, fluid mechanics